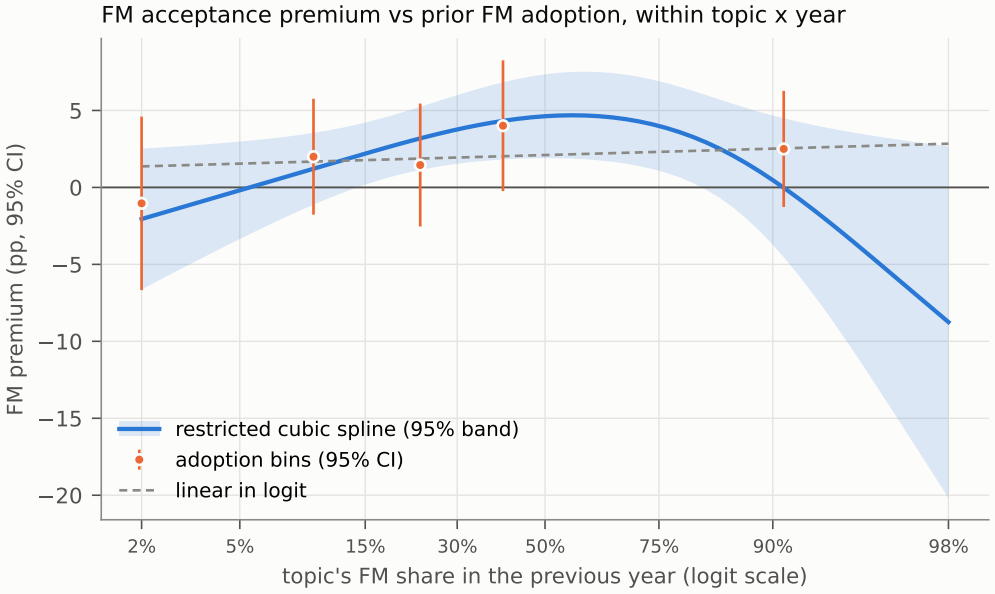PH5 fmwave (post hoc, exploratory; decided papers 2023-2026)